

Condition-Wise Sinkhorn Drifting for One-Shot Learned Channel Simulation

Rick Fritschek and Rafael F. Schaefer

Abstract—Learned communication systems may evaluate stochastic channel surrogates millions of times inside differentiable training loops, making diffusion-style reverse sampling expensive. This paper proposes condition-wise Sinkhorn drifting, a one-shot channel surrogate that preserves the transmitted symbol and transports only the conditional output laws $p(y | x)$. We formulate a conditional Sinkhorn objective over repeated outputs at the same transmitted symbol and train the generator with finite-sample barycentric velocities followed by detached particle regression. Experiments on additive white Gaussian noise (AWGN), Rayleigh fading, solid-state power amplifier (SSPA) nonlinearity, and a compact tapped-delay-line (TDL) channel compare direct drifting, joint Sinkhorn drifting, condition-wise Sinkhorn drifting, conditional denoising diffusion probabilistic modeling (DDPM), denoising diffusion implicit modeling (DDIM), and Wasserstein generative adversarial network (WGAN) references. Within the evaluated one-shot drifting-family variants, condition-wise Sinkhorn is strongest under conditional diagnostics and symbolic-coding checks, while diffusion remains strongest on the hardest downstream symbol-error-rate (SER) curves. The resulting operating point is a condition-preserving one-shot simulator for settings where repeated channel calls make diffusion-style sampling too costly.

Index Terms—learned channel simulation, conditional generative models, Sinkhorn divergence, diffusion models, inference latency

I. INTRODUCTION

Accurate channel models are a basic requirement for modern communication system design. They are particularly important in neural and end-to-end learned systems, where the channel appears repeatedly inside training loops, decoder adaptation procedures, and differentiable autoencoder pipelines. When a clean analytical channel model is unavailable, too restrictive, or poorly matched to measured data, generative channel surrogates become attractive because they can learn the full conditional distribution directly from samples.

This problem also involves more than distributional accuracy. In many learning pipelines, the channel model is evaluated millions of times. As a result, sample quality, conditioning accuracy, computational cost, and differentiability all matter simultaneously. The practical question is whether a generative model can match the channel distribution at a cost compatible with large-scale training and deployment workflows.

Recent work on diffusion-based channel modeling has shown that conditional diffusion models can generate high-quality channel samples across standard benchmarks such as additive white Gaussian noise (AWGN), Rayleigh fading, and nonlinear amplifier channels [1]. Follow-up work has further

studied robustness and speed-quality tradeoffs for diffusion-based channel generation [2], extended diffusion-based channel synthesis to high-dimensional user-specific channels [3], and proposed diffusion-driven digital-twin-style generation of statistical channel state information from sensing and location information [4]. More broadly, diffusion methods have also begun to influence adjacent wireless tasks such as high-dimensional channel estimation [5], low-complexity multiple-input multiple-output (MIMO) channel estimation [6], [7], joint channel estimation and data detection [8], and fast time-varying MIMO orthogonal frequency-division multiplexing (MIMO-OFDM) estimation [9]. This line of work is compelling because diffusion models are flexible, robust, and often more stable than earlier adversarial approaches. This improvement comes with a practical system cost. Inference requires an iterative reverse denoising procedure, and the latency scales with the number of diffusion steps. This tradeoff is manageable when the channel generator is called occasionally. It becomes more problematic when the generator is embedded in inner loops of iterative training or repeated Monte Carlo evaluations. This latency bottleneck has motivated several acceleration strategies for denoising diffusion probabilistic modeling [10], including skipped or reduced-step sampling as in denoising diffusion implicit modeling (DDIM), as well as one-step or few-step alternatives such as consistency models and distillation-based approaches that compress multi-step samplers into substantially shorter generation paths [11]–[13]. This motivates the study of one-shot conditional generators for learned channel simulation. Among recently proposed generative paradigms, drifting models [14] are particularly interesting because they learn the transport from a simple latent source to the target data distribution directly in the network parameters. Intuitively, diffusion spends computation at test time by progressively transforming noise into data through a reverse-time chain. Drifting spends computation during training. Generated samples are repeatedly nudged toward the data manifold by a drift field, and the neural generator is updated so that future samples move closer to the desired distribution in a single forward pass. Once trained, sampling is therefore naturally one-shot. Here, one-shot means one neural generator evaluation per channel sample, with the usual latent draw and conditioning input but no reverse denoising chain or iterative particle correction at inference. This perspective is also consistent with recent geometric analyses of generative dynamics that relate diffusion-like behavior to autonomous vector-field structure and marginalized noise geometry [15]. Recent work on one-step generation via Wasserstein gradient flows (W-Flow) [16] sharpens this viewpoint by replacing heuristic drifting fields with Sinkhorn-divergence optimal-

transport velocities. This is directly relevant to channel simulation, because the quality and stability of the drift field is the main design question once inference is constrained to one generator evaluation.

The distinction between ordinary generation and channel simulation is central. In ordinary generative modeling, samples are transported in the data space until the model distribution matches the target distribution. In channel simulation, the transmitted symbol is fixed side information. The relevant target is the family of conditional laws $\{p(\cdot | x)\}_{x \sim \mu}$. A generator can match the global output cloud while mixing samples from different transmitted symbols, which produces an incorrect channel surrogate for communication-system training. This motivates a condition-wise transport formulation in which couplings preserve x and move only the output component y . The proposed condition-wise Sinkhorn drifting implements this idea as a one-shot neural channel generator. It uses drifting's detached-target regression and the Sinkhorn/W-Flow barycentric velocity as building blocks. The channel-specific contribution is to constrain the transport problem to repeated outputs at the same transmitted symbol, so that the learned surrogate moves samples inside $p(y | x)$ rather than across different transmitted symbols.

The emphasis of this study is on generator-level distributional accuracy, computational tradeoffs, and downstream channel-implant checks. A full communication-system design study over all modulation and coding regimes is outside the present scope. The condition-wise Sinkhorn estimator also assumes simulator-style access to repeated outputs at the same transmitted symbol. Measured datasets with only one observation per condition require local conditional approximations, which we formulate only as a limitation and future extension.

The main contributions are summarized as follows.

- We formulate condition-wise Sinkhorn drifting for learned channel simulation by integrating Sinkhorn divergences over the fixed-input conditional laws $p(\cdot | x)$.
- We connect this objective to a conditional Wasserstein-flow velocity, then state how the practical method differs from the population flow through finite-sample barycentric estimation and projected detached-target neural training.
- We implement direct-output conditional drifting and compare joint/global Sinkhorn transport with condition-wise Sinkhorn transport under a matched one-shot generator.
- We compare the proposed one-shot surrogate against conditional DDPM, conditional DDIM, and conditional WGAN reference models on AWGN, Rayleigh fading, SSPA, and compact TDL channel studies wherever the protocols overlap.
- We evaluate channel surrogates with direct SWD, anchor-conditioned moment metrics, downstream SER, bit error rate (BER), and timing measurements, showing that communication usefulness is better reflected by conditional and downstream metrics than by global SWD alone.

Source code will be made available through the first author's GitHub repository upon publication¹.

¹https://github.com/Fritschek/conditional_drifting_models.

II. PROBLEM FORMULATION AND DRIFTING BACKGROUND

A. What drifting changes conceptually

Because drifting is very recent and remains less familiar than diffusion, we briefly summarize its generative viewpoint. Following [14], let f_θ map a latent variable $z \sim p_z$ to a sample $u = f_\theta(z)$, and let

$$q_\theta = (f_\theta)_\# p_z \quad (1)$$

denote the pushforward distribution induced by the generator. The training objective uses a *drifting field* that tells us how samples drawn from the current model distribution should move so that the model distribution approaches the target distribution p .

At a high level, the desired property is an equilibrium condition. If the model already matches the target distribution, then the drift should vanish. Using $V_{p,q}(u)$ to denote the field evaluated at a sample u under target distribution p and model distribution q , the fixed-point intuition can be summarized as

$$q = p \implies V_{p,q}(u) = 0, \quad (2)$$

which is the sense in which matched distributions form an equilibrium of the training dynamics [14].

This makes the training logic very different from diffusion. In diffusion, one learns a denoiser and then performs iterative refinement at inference time. In drifting, one instead constructs improved targets during training by moving current samples along the field $V_{p,q}$, and then regresses the generator onto those improved targets. As a result, the iterative correction happens during optimization.

The one-step view can be written as follows. Starting from a generated sample $u = f_\theta(z)$, define a drifted target

$$\tilde{u} = \text{stopgrad}(u + \eta V_{p,q_\theta}(u)), \quad (3)$$

and optimize the generator by

$$\mathcal{L}_{\text{drift}} = \mathbb{E}_{z \sim p_z} [\|f_\theta(z) - \tilde{u}\|_2^2]. \quad (4)$$

The stop-gradient is essential here because it makes the drifted sample behave like a fixed target for the current optimization step. Repeated training steps therefore evolve the pushforward q_θ toward the target distribution. This is also the cleanest way to see why drifting is naturally one-shot at inference. Once the generator has learned to emit samples near equilibrium, sampling only requires drawing z and evaluating $f_\theta(z)$, without a reverse-time chain.

The W-Flow formulation [16] connects this one-shot training principle to entropic optimal transport. Let $\text{OT}_\varepsilon(q, p)$ denote the entropic optimal-transport cost between a generated law q and target law p , computed with Sinkhorn iterations [17]. The debiased Sinkhorn divergence is

$$S_\varepsilon(q, p) = \text{OT}_\varepsilon(q, p) - \frac{1}{2} \text{OT}_\varepsilon(q, q) - \frac{1}{2} \text{OT}_\varepsilon(p, p), \quad (5)$$

which removes the entropic self-bias and interpolates between optimal-transport and kernel-like behavior as ε changes [18].

For an empirical generated particle u , the corresponding W-Flow velocity can be written in barycentric-projection form as

$$V_\varepsilon(u) = T_{q,p}^\varepsilon(u) - T_{q,q}^\varepsilon(u), \quad (6)$$

where $T_{q,p}^\varepsilon$ maps a generated sample to the barycenter of target samples under the entropic optimal-transport coupling, and $T_{q,q}^\varepsilon$ is the generated self-transport projection. This has the same attraction–repulsion structure as the practical drifting rule. The weights come from a globally mass-constrained coupling that replaces independently row-normalized kernels.

B. Conditional channel simulation

We now specialize the generic drifting view to learned conditional channel simulation. Let $x \in \mathbb{R}^n$ denote the transmitted channel input and $y \in \mathbb{R}^n$ the channel output. The objective is to learn a conditional generator that reproduces the channel law given x . In this setting, conditioning enters through the transmitted symbol, while stochasticity is carried by a latent variable $z \sim \mathcal{N}(0, I)$.

The experiments use a direct channel-output formulation. The generator approximates $p(y | x)$ and predicts

$$\hat{y} = g_\theta(x, z). \quad (7)$$

This choice aligns the drifting generator with the direct-output diffusion and WGAN baselines and with the downstream channel-implant evaluation, where the learned surrogate is used as a simulator for y given x . Residual-output variants were examined during development and left out of the final benchmark after weaker direct-output and downstream channel-implant performance.

C. Conditional drifting model

We use a practical conditional adaptation inspired by [14]. To avoid confusion with the communication input symbol x , we use u above for the generic sample variable in the abstract formulation and reserve x below for the transmitted symbol. In our implementation, the generator predicts $\hat{y} = g_\theta(x, z)$, where

$$z \sim \mathcal{N}(0, I). \quad (8)$$

The network is implemented as a two-hidden-layer multilayer perceptron with hidden width 128, latent dimension 16, and SiLU activations.

$$[x, z] \rightarrow \text{Linear} \rightarrow \text{SiLU} \rightarrow \text{Linear} \rightarrow \text{SiLU} \rightarrow \text{Linear}. \quad (9)$$

The training target is formed from an attraction–repulsion drift field in direct output space. This is a concrete channel-specific instantiation of the generic field $V_{p,q}$. Given generated samples $G = \{g_i\}$ and positive samples $P = \{p_j\}$, we compute

$$V(g_i) = \left(\sum_j w_{ij}^{(P)} p_j - g_i \right) - \lambda \left(\sum_k w_{ik}^{(G)} g_k - g_i \right), \quad (10)$$

where $w^{(P)}$ and $w^{(G)}$ are row-normalized RBF kernel weights. The first term attracts each generated sample toward a local barycenter of true output samples, and the second term repels it

from nearby generated samples to reduce collapse. This yields a detached target

$$\tilde{g}_i = \text{stopgrad}(g_i + \eta V(g_i)), \quad (11)$$

and the generator is optimized by

$$\mathcal{L}_{\text{drift}} = \frac{1}{|G|} \sum_i \|g_i - \tilde{g}_i\|_2^2. \quad (12)$$

The generic update $u \mapsto u + \eta V_{p,q}(u)$ from the drifting formulation is realized here with a kernel-based drift field estimated directly from minibatches of true and generated output samples. This empirical field acts as a kernel particle interaction rule. The attraction term pulls generated samples toward the target sample cloud, while the repulsion term prevents generated particles from collapsing onto a few modes. In this sense, the method is related in spirit to kernel-based distribution shaping and MMD-like matching. Here it is used as a practical training rule, without deriving a formal discrepancy minimization objective.

In the simulations, the drift scale is $\eta = 1.0$, the repulsive weight is $\lambda = 1.0$, the minimum kernel bandwidth is 0.2, and the maximum drift norm is 2.0. These settings were chosen because they produced stable training behavior and materially improved the early AWGN drifting baselines.

a) Conditioning-aware kernel variant. For the enhanced direct-output drifting model, we replace the target-only kernel by a kernel on joint condition-target pairs. Instead of computing sample interactions from $k(y_i, y_j)$, we compute them from $k((x_i, y_i), (x_j, y_j))$, so that attraction and repulsion depend on similarity in both the channel input and the output space. This aligns the drift field more closely with the conditional channel structure. The generator architecture is unchanged.

This kernel rule is the conditioning-aware baseline used for the direct-output drifting variants. It is computationally simple. Its main open design choice is the minibatch coupling used to build V , which remains channel-specific. We therefore also study Sinkhorn-based drift fields that replace the row-normalized kernel interaction with the velocity in (6), while preserving the same detached-target training loop and the same one-shot inference path.

D. Condition-wise Sinkhorn drifting

The W-Flow objective in [16] is unconditional. Both p and q_θ are distributions on the generated sample space. For channel simulation, the generated and true joint laws share the same transmitted-symbol marginal. Writing μ for the input distribution,

$$p(dx, dy) = \mu(dx)p_x(dy), \quad q_\theta(dx, dy) = \mu(dx)q_{\theta,x}(dy). \quad (13)$$

The condition x is fixed side information. Transporting in the full joint space (x, y) therefore solves a different problem from matching the channel law $p(y | x)$, because it allows the geometry of the fixed condition variable to influence the coupling while the update can only move the output component.

We instead define the conditional Sinkhorn functional

$$\mathcal{S}_\varepsilon^{\text{cond}}(q_\theta, p) = \int S_\varepsilon(q_\theta, p_x) \mu(dx), \quad (14)$$

where the Sinkhorn divergence is computed separately over output samples for each fixed transmitted symbol. For brevity, we call the output cloud at a fixed transmitted symbol an output fiber. In measure-theoretic terms, $p_x = p(\cdot | x)$ is the conditional measure over that fiber. Throughout the paper we refer to the resulting estimator as condition-wise Sinkhorn drifting. Equivalently, admissible couplings must preserve the condition.

$$\pi(dx, dy, d\bar{y}) = \mu(dx) \pi_x(dy, d\bar{y}), \quad \pi_x \in \Pi(q_{\theta, x}, p_x). \quad (15)$$

The Wasserstein gradient flow of (14) has zero x -velocity and evolves only inside each fixed-input conditional law.

$$\partial_t q_t(x, y) + \nabla_y \cdot (q_t(x, y) v_t(x, y)) = 0. \quad (16)$$

Applying the W-Flow velocity condition-wise gives

$$v_t(x, y) = T_{q_t, x, p_x}^\varepsilon(y) - T_{q_t, x, q_t, x}^\varepsilon(y). \quad (17)$$

For channel simulation, the key consequence is the following equilibrium statement.

Proposition 1 (Condition-wise Sinkhorn equilibrium): Under the standard Sinkhorn-divergence assumptions of [18], the conditional objective $\mathcal{S}_\varepsilon^{\text{cond}}$ is nonnegative and is zero exactly when

$$q_{t, x} = p_x \quad \text{for } \mu\text{-a.e. } x. \quad (18)$$

Along the corresponding population W-Flow dynamics,

$$\frac{d}{dt} \mathcal{S}_\varepsilon^{\text{cond}}(q_t, p) = - \int \|v_t(x, \cdot)\|_{L^2(q_t, x)}^2 \mu(dx) \leq 0. \quad (19)$$

Thus the equilibrium of the condition-wise flow is equality of the full family of channel laws $p(\cdot | x)$.

Proof: The zero-set statement follows by applying the nonnegativity and unique-zero property of the Sinkhorn divergence to each fixed x and integrating over μ . The derivative identity is the same fixed-condition W-Flow identity, again integrated over μ . $\square$

The proposition gives the population conditional optimal-transport picture needed for channel simulation: equality is enforced for almost every fixed input. Matching a joint sample cloud under an arbitrary joint cost does not enforce this condition-wise equality.

The implemented algorithm approximates this population object in two ways. First, it estimates the barycentric velocity from finite minibatch Sinkhorn couplings. Second, it projects the explicit particle update back into the neural generator class through detached-target regression. The resulting update differs from exact gradient descent on $\mathcal{S}_\varepsilon^{\text{cond}}$ with respect to θ . The population conditional flow therefore serves as a fixed-point guide for the projected training rule. If the generator family cannot realize all conditional laws p_x , the practical fixed point is the best representable conditional surrogate reached by this projected training dynamics. For anchors $x_i \sim \mu$, the estimator draws multiple generated samples $\hat{y}_{i, k} = g_\theta(x_i, z_{i, k})$, multiple positive channel samples

TABLE I
CONDITION-WISE SINKHORN DRIFTING UPDATE.

Input: generator g_θ , channel sampler $p(\cdot x)$, anchors $\{x_i\}_{i=1}^B$, generated count K_g , positive count K_p , reference count K_r , drift scale η .
1. Draw $\hat{y}_{i, k} = g_\theta(x_i, z_{i, k})$, $k = 1, \dots, K_g$.
2. Draw positive samples $y_{i, j} \sim p(\cdot x_i)$, $j = 1, \dots, K_p$.
3. Draw generated reference samples $\hat{y}'_{i, \ell} = g_\theta(x_i, z'_{i, \ell})$, $\ell = 1, \dots, K_r$.
4. For each anchor i , solve two output-space Sinkhorn problems, $\hat{Q}_i \leftrightarrow P_i$ and $\hat{Q}_i \leftrightarrow \hat{Q}'_i$.
5. Form barycentric velocity $v_{i, k} = T_{\hat{Q}_i, P_i}^\varepsilon(\hat{y}_{i, k}) - T_{\hat{Q}_i, \hat{Q}'_i}^\varepsilon(\hat{y}_{i, k})$.
6. Regress $g_\theta(x_i, z_{i, k})$ toward $\text{stopgrad}(\hat{y}_{i, k} + \eta v_{i, k})$.

$y_{i, j} \sim p(\cdot | x_i)$, and an independent generated reference batch $\hat{y}'_{i, \ell} = g_\theta(x_i, z'_{i, \ell})$. Sinkhorn barycentric projections are then computed separately for each anchor:

$$v_{i, k} = T_{\hat{Q}_i, P_i}^\varepsilon(\hat{y}_{i, k}) - T_{\hat{Q}_i, \hat{Q}'_i}^\varepsilon(\hat{y}_{i, k}), \quad (20)$$

followed by the detached regression target

$$\tilde{y}_{i, k} = \text{stopgrad}(\hat{y}_{i, k} + \eta v_{i, k}). \quad (21)$$

Table I summarizes the resulting training update.

a) Local conditional approximation.: The exact estimator above is natural for simulator channels where repeated draws at the same x_i are available. Measured datasets often contain only one or a few observations per condition. In that setting, the same geometry can be approximated by replacing the pointwise conditional law with a local kernel estimate in condition space. Given measured pairs $\{(x_r, y_r)\}_{r=1}^N$, define

$$\alpha_r(x_0) = \frac{K_h(x_r, x_0)}{\sum_{\ell=1}^N K_h(x_\ell, x_0)}, \quad \hat{p}_{h, x_0} = \sum_{r=1}^N \alpha_r(x_0) \delta_{y_r}, \quad (22)$$

where K_h is a bandwidth-controlled kernel on transmitted symbols or channel-state features. For a generated batch at anchor x_0 , condition-wise Sinkhorn is then replaced by a weighted local Sinkhorn problem between $\{\hat{y}_k = g_\theta(x_0, z_k)\}$ and $\hat{p}_{h, x_0}$. Equivalently, a generated local reference law

$$\hat{q}_{\theta, h, x_0} = \sum_{r=1}^N \alpha_r(x_0) \frac{1}{K} \sum_{k=1}^K \delta_{g_\theta(x_r, z_{r, k})} \quad (23)$$

can be used when the learned generator should match the same local conditioning bandwidth as the measured target estimate. The bandwidth h controls the usual bias–variance tradeoff: small neighborhoods preserve conditioning but increase estimator variance, while large neighborhoods reduce variance but approach a global transport problem. The experiments in this paper use simulator access and therefore the exact repeated-condition estimator. The local formulation is included only as a measured-data direction. Bandwidth selection and neighborhood bias remain open.

The major operational advantage is straightforward. Once training is complete, sampling requires only a single evaluation of $g_\theta(x, z)$, which produces $\hat{y}$ directly. Sampling avoids reverse chains, ODE solvers, and per-sample iterative schedules at inference.

III. EXPERIMENTAL SETUP

A. Channel models

We use four stochastic channel families. All channels are implemented as real-valued maps $x \mapsto y$, but the coordinate interpretation differs by channel. AWGN and Rayleigh use real-valued vectors directly and have dimension $n = 7$ in the generator-level benchmark. SSPA and TDL use paired in-phase/quadrature coordinates with $n = 8$, corresponding to four complex symbols. Noise variables are sampled independently across real coordinates unless the deterministic channel transformation or tapped-delay convolution introduces coupling. The generator-level reference comparison is reported on AWGN, Rayleigh, and SSPA, where matched DDPM, DDIM, WGAN, and direct-drifting reference runs are available. TDL is included in the Sinkhorn drift-field ablation and downstream channel-implant study as a compact wireless channel with memory. For TDL, the DDPM, DDIM, and WGAN direct-output SWD rows are evaluated from the trained baseline checkpoints used in the SER-curve study.

AWGN:

$$y = x + n, \quad n \sim \mathcal{N}(0, \sigma^2 I). \quad (24)$$

This is the standard additive white Gaussian noise reference channel and serves as the simplest near-identity baseline.

Rayleigh: the transmitted symbol is multiplied elementwise by a random fading amplitude and then corrupted by additive Gaussian noise:

$$y = h \odot x + n, \quad (25)$$

where $\odot$ denotes elementwise multiplication,

$$h_k = \frac{1}{\sqrt{2}} \sqrt{u_k^2 + v_k^2}, \quad u_k, v_k \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1), \quad (26)$$

and $n \sim \mathcal{N}(0, \sigma^2 I)$.

Each real component is therefore scaled by an independent Rayleigh-distributed magnitude before noise is added. The downstream encoder and decoder are not given h_k , and no explicit equalization is applied. As implemented here, Rayleigh is a real-valued vector channel. The paired-I/Q complex representation is used for SSPA and TDL.

SSPA: the input is passed through a smooth solid-state power amplifier nonlinearity before additive noise is applied. In this case, the real-valued representation is interpreted as paired I/Q components. Let

$$r = \sqrt{x_I^2 + x_Q^2} \quad (27)$$

denote the input amplitude. The multiplicative gain factor used in the benchmark is

$$a_{\text{SSPA}}(r) = \frac{G}{\left(1 + \left(\frac{Gr}{A_0}\right)^{2p}\right)^{1/(2p)}}, \quad (28)$$

with default parameters $p = 3$, $A_0 = 1.5$, and $G = 5.0$. Here $a_{\text{SSPA}}(r)$ is a gain, not the output amplitude. The corresponding output amplitude is

$$A_{\text{out}}(r) = a_{\text{SSPA}}(r)r = \frac{Gr}{\left(1 + \left(\frac{Gr}{A_0}\right)^{2p}\right)^{1/(2p)}}, \quad (29)$$

which is the Rapp-style AM/AM law used in the benchmark. The noiseless amplifier output is $\tilde{x} = a_{\text{SSPA}}(r)x$, and the observed channel output is

$$y = \tilde{x} + n, \quad n \sim \mathcal{N}\left(0, \frac{\sigma^2}{2} I\right). \quad (30)$$

This channel introduces amplitude-dependent nonlinear distortion while preserving the input direction in the I/Q plane.

TDL: The TDL channel is a compact short-block approximation of the 3GPP TR 38.901 TDL-D profile [19]. The real vector is interpreted as a complex codeword $s_m = x_{2m} + jx_{2m+1}$, $m = 0, \dots, N_c - 1$, with $N_c = 4$ complex channel uses in the symbolic coding runs. For each codeword, the channel samples a finite set of complex taps and applies circular convolution,

$$\tilde{s}_m = \sum_{\ell=0}^{L_h-1} h_\ell s_{(m-d_\ell) \bmod N_c}. \quad (31)$$

The observed output is

$$y_m = \tilde{s}_m + n_m, \quad n_m \sim \mathcal{CN}(0, \sigma^2). \quad (32)$$

The implemented profile uses the TDL-D line-of-sight structure with first-tap K -factor 13.3 dB. Because the codeword is short, the normalized path delays from the standard profile are rounded to a small set of effective circular shifts and path powers are renormalized after aggregation. This keeps the channel compatible with the short symbolic autoencoder while preserving the random multipath mixture structure. We use this TDL configuration as a compact channel-with-memory stress test. It is intentionally smaller than a full OFDM or long-block TDL evaluation.

B. Training protocol

The experiments are organized into two complementary protocols. First, the generator-level protocol compares the proposed drifting family against the established conditional DDPM, DDIM, and WGAN references on AWGN, Rayleigh, and SSPA. TDL is included in the final reported benchmark through the Sinkhorn drift-field ablation, the direct-SWD comparison, and the downstream SER curves. Second, the drift-field ablation holds the drifting network fixed and compares global/joint Sinkhorn drifting with condition-wise Sinkhorn drifting. Target-space and conditioning-aware kernel drifts were run as supporting controls, but the main tables keep the transport comparison focused. This second protocol is reported over 100 seeds for the main drift-field runs and is evaluated with direct SWD, anchor-conditioned metrics, and downstream symbolic SER/BER. It is the controlled comparison in this paper because the generator architecture, optimizer family, and one-shot inference path are held fixed across drift fields. The DDPM, DDIM, and WGAN rows provide best-effort reference baselines using stable model-family settings. A matched-capacity ranking against drifting would require an additional equal-parameter and equal-budget study. For SSPA, we additionally report a 30-seed update-budget-controlled condition-wise Sinkhorn operating point because the corrected same-condition transport field saturates after far

fewer optimizer updates than the long diffusion-style SSPA preset. The selected operating point follows the observed W-Flow training behavior and the anchor-conditioned diagnostics. The TDL channel is included in the drift-field study and in the downstream SER curves with diffusion and WGAN references. For the direct-output SWD table, TDL W-Flow rows come from the drift-field ablation, and TDL DDPM/DDIM/WGAN rows are obtained by checkpoint-only direct-SWD evaluation of the corresponding trained baseline implants.

a) *Channel and data settings.*: Table II collects the channel dimensions, message alphabets, training points, and generator-evaluation budgets. The noise standard deviations induced by these $(E_b/N_0, R)$ pairs are 0.5260, 0.2350, 0.3251, and 0.3162 for AWGN, Rayleigh, SSPA, and TDL. The SSPA condition-wise Sinkhorn row uses the compact update-budget-controlled operating point because the corrected same-condition field had already converged at this budget. Table VII reports the corresponding SSPA budget sensitivity: early budgets remain undertrained, the compact row is close to the analytic anchor floors, and the full 10^7 -sample preset worsens both anchor diagnostics and downstream SER for this sharper field. Downstream symbolic autoencoders are trained through either the analytic channel or a learned channel implant and are always evaluated on the analytic channel. Within each channel, the autoencoder architecture, optimizer schedule, message alphabet, rate, training E_b/N_0 , and epoch budget are fixed; only the channel implant used during training changes. Reported downstream uncertainty is taken over the complete seeded training and evaluation pipeline.

b) *Model settings.*: Drifting uses a conditional MLP with latent dimension 16, hidden width 128, two hidden layers, SiLU activations, and Adam with learning rate 10^{-3} . The base drifting objective uses drift scale 1.0, median-heuristic bandwidth selection, minimum bandwidth 0.2, maximum drift norm 2.0, and repulsive weight 1.0. The Sinkhorn drift-field variants use the same direct-output generator architecture and detached-target update, but replace the training drift field. The joint Sinkhorn row uses a Sinkhorn/W-Flow drift in joint condition-output features, while the condition-wise row uses the per-anchor conditional Sinkhorn field in (17). Unless otherwise stated, Sinkhorn variants use 10 Sinkhorn iterations, automatic entropic scale selection with minimum $\varepsilon = 10^{-3}$, four generated samples, four positive channel samples, and four generated reference samples per condition anchor.

For AWGN, Rayleigh, and SSPA, diffusion models $p(y | x)$ directly with v -prediction, cosine-ZF variance schedule, 100 diffusion steps, and EMA decay 0.9. Hidden widths are 110 (AWGN), 128 (Rayleigh), and 110 (SSPA). AWGN and Rayleigh use a two-stage learning-rate schedule (10 epochs at 10^{-3} , then 20 epochs at 10^{-4}), while SSPA uses a constant 10^{-4} for 160 epochs. We report DDPM and DDIM with 100, 50, 20, and 10 denoising steps.

WGAN uses a conditional Wasserstein objective with RM-Sprop at 10^{-4} for generator and critic, five critic updates per generator update, and weight clipping at 0.01. Hidden widths are 128 (AWGN) and 256 (Rayleigh/SSPA). WGAN is reported in direct output space wherever it is compared against direct-output channel generators.

c) *Experimental fairness.*: The benchmark is intended as a practical comparative study, with stable model-family implementations prioritized over strict matched-capacity control. Architectures and training schedules were chosen as stable implementations for each model family. Identical parameter counts and compute budgets across drifting, diffusion, and WGAN are outside the benchmark design. The Sinkhorn rows are therefore reported as a drift-field ablation under a matched drifting architecture and training budget. They are compared to diffusion and WGAN wherever the observable metric and channel protocol overlap. The downstream W-Flow table focuses on the learned drifting/Sinkhorn surrogates and the analytic-channel reference training condition. The SER curves add WGAN and DDIM-100 when all corresponding channel implants are available. The central ablation is the drift-field geometry: joint/global transport versus condition-wise Sinkhorn transport under the same one-shot generator.

As an additional diagnostic, Table III reports inference-network parameter counts used in this benchmark. The Sinkhorn rows have the same parameter count as direct drifting because the transport computation changes the training update, not the deployed generator.

d) *Seeding and reporting.*: The diffusion/WGAN reference benchmark results are reported as mean $\pm$ standard deviation over ten seeds. The Sinkhorn drift-field ablation is reported over 100 seeds unless stated otherwise. The SSPA update-budget check uses 30 seeds. We use standard deviation for the ten-seed reference rows to describe run-to-run variability. W-Flow tables use mean $\pm$ standard error because the main purpose of those rows is to compare seed-averaged drift-field variants. Each run uses deterministic initialization to reduce run-to-run variance unrelated to model stochasticity.

C. Evaluation metric

The diffusion/WGAN/drifting reference comparison emphasizes generator-level distributional accuracy and timing, while the Sinkhorn drift-field ablation adds anchor-conditioned diagnostics and downstream symbolic SER/BER. The primary generator metric is direct-output sliced Wasserstein distance (SWD). For generated channel outputs, we report $\text{SWD}(\{\hat{y}_i\}, \{y_i\})$. Given true samples $P = \{v_i\}_{i=1}^N$ and generated samples $Q = \{\hat{v}_i\}_{i=1}^N$, SWD is estimated by drawing random unit directions θ_m , projecting both clouds onto those directions, computing the one-dimensional Wasserstein distance for each projection, and averaging over projections:

$$\text{SWD}(P, Q) = \frac{1}{M} \sum_{m=1}^M W_1(\{\langle \theta_m, v_i \rangle\}_{i=1}^N, \{\langle \theta_m, \hat{v}_i \rangle\}_{i=1}^N).$$

Thus SWD compares the generated and true sample clouds through many one-dimensional views and captures geometric mismatch more faithfully than a purely bin-based marginal comparison. In our experiments, SWD is estimated using 128 random one-dimensional projections on AWGN, Rayleigh, SSPA, and TDL. The projection directions are sampled from an isotropic Gaussian distribution, normalized to unit length, and generated from the deterministic metric seed associated with the run. SWD is used primarily for comparability with

TABLE II

SIMULATION SETUP USED FOR THE JOURNAL EXPERIMENTS. THE LISTED E_b/N_0 VALUES ARE THE NOMINAL CHANNEL-MODEL AND AUTOENCODER TRAINING POINTS. GENERATOR-LEVEL AWGN/RAYLEIGH/SSPA REFERENCE ROWS USE TEN SEEDS. W-FLOW ROWS USE 100 SEEDS EXCEPT FOR THE 30-SEED SSPA UPDATE-BUDGET-CONTROLLED OPERATING POINT. ALL SWD EVALUATIONS USE 128 RANDOM PROJECTIONS.

Channel	n	M_{msg}	R	E_b/N_0	Train samples / batch / epochs	Eval. samples	Main use
AWGN	7	16	4/7	5 dB	$10^7/5000/30$	10^6	Reference + W-Flow
Rayleigh	7	16	4/7	12 dB	$10^7/5000/30$	10^6	Reference + W-Flow
SSPA reference	8	64	6/8	8 dB	$10^7/4096/160$	10^6	DDPM/DDIM/WGAN + direct drifting
SSPA W-Flow operating point	8	64	6/8	8 dB	$1.2 \times 10^5/4096/160$	10^6	Reported condition-wise Sinkhorn
TDL	8	16	4/8	10 dB	$1.2 \times 10^5/512/60$	10^5	W-Flow + downstream curves

TABLE III

MODEL PARAMETER COUNTS IN THE BENCHMARK SETUP. COUNTS REFER TO INFERENCE-TIME NETWORKS. SINKHORN DRIFTING USES THE SAME ONE-SHOT GENERATOR AS DIRECT DRIFTING. ITS SINKHORN COUPLINGS ARE TRAINING-TIME COMPUTATIONS AND ADD NO INFERENCE-TIME PARAMETERS. DDPM AND DDIM SHARE THE SAME DENOISER WEIGHTS. WGAN ROWS REPORT GENERATOR-SIDE PARAMETERS.

Model	AWGN	Rayleigh	SSPA	TDL
Direct drifting	20,487	20,487	20,744	20,744
Sinkhorn drifting variants	20,487	20,487	20,744	20,744
Diffusion denoiser (DDPM / DDIM)	59,847	74,247	60,178	74,632
WGAN generator	19,335	71,431	72,200	72,200

prior diffusion-channel benchmarks [1], [2]. Compared with histogram-based L_1 distance, it also better captures sample geometry in $\mathbb{R}^n$. SWD provides a geometry-aware comparison of global output sample clouds. It leaves decision-relevant behavior such as error rates unmeasured. Those depend on the joint structure of (x, y) . For communication use, two generators with similar global sample-cloud discrepancy can induce different conditional means, conditional covariance errors, and downstream symbol-error rates. The Sinkhorn drift-field study therefore also includes anchor-conditioned moment metrics and symbolic channel-implant SER/BER checks when evaluating the proposed drifting variants. We therefore report both downstream-dependent and generator-level metrics. Downstream SER and BER under analytic-channel evaluation measure the utility of a surrogate inside the trained communication system, but they also depend on the chosen encoder, decoder, optimizer, and training point. SWD is independent of a downstream receiver and remains useful for comparing output distributions across generators, while global SWD can miss fixed-input conditional errors. Anchor-conditioned metrics bridge these views by probing repeated outputs at fixed transmitted symbols.

IV. BENCHMARK RESULTS

A. Unified SWD benchmark across channels

Table IV reports a unified direct-output SWD comparison across AWGN, Rayleigh, SSPA, and TDL. The reference rows summarize the diffusion and WGAN baselines. The drifting-family rows contain direct drifting and the two Sinkhorn drift fields under a matched one-shot architecture. The controlled conclusion from this table is the ranking inside the drifting-family block, while the diffusion and WGAN rows provide reference context from their own stable training protocols.

DDIM-100 gives the lowest direct-space SWD on AWGN, Rayleigh, and SSPA among the matched reference rows. Among the Sinkhorn variants, condition-wise Sinkhorn is strongest under direct SWD on AWGN, Rayleigh, and TDL, and is numerically close to direct drifting on SSPA under the compact update-budget run. Within the full drifting-family block, condition-wise Sinkhorn is lowest on AWGN, Rayleigh, and TDL, while direct drifting remains lowest on SSPA under this global metric. The drift-field ranking is therefore channel-dependent.

Timing follows the expected pattern. Drifting and WGAN operate at roughly $0.1 \mu\text{s}$ per sample on the local CUDA timing setup, while DDPM and DDIM require microsecond-to-tens-of-microseconds sampling times depending on the sampler and the number of denoising steps. Thus, even when diffusion achieves the strongest SWD, one-shot generators retain a substantial practical latency advantage. The W-Flow variants retain the same one-shot inference path as direct drifting. Their cost differences are training-time drift-field costs. Table V therefore reports W-Flow ablation rows under the same 2% local CUDA timing protocol. The timing run uses PyTorch on a local NVIDIA RTX 5060 Ti GPU in standard float32 execution, without CUDA graph capture, `torch.compile`, or mixed precision. Training time is measured by running the actual training loop on 2% of the configured dataset size, rounded to full batches, and extrapolating by the ratio of timed optimizer steps to full optimizer steps. This includes synthetic channel sampling, forward and backward passes, optimizer updates, and the Sinkhorn computations used by W-Flow rows. It excludes checkpoint writing and downstream symbolic autoencoder training. Inference timing uses batch size 512, 40 batches per repeat, two warmup repeats, seven timed repeats, and CUDA synchronization around each repeat. It includes drawing the conditioning inputs and evaluating the generator. For diffusion rows, inference includes the complete DDPM or DDIM denoising trajectory and the same conditioning overhead. The total column adds projected training time to the time needed to generate the evaluation-sample budget listed in Table II. On AWGN, Rayleigh, and SSPA, the condition-wise Sinkhorn row is faster to train than joint Sinkhorn because it solves many small same-condition transport problems instead of forming a global minibatch interaction matrix. This agrees with the leading coupling cost: with B anchors, K_g generated particles, K_p positives, K_r generated references, and I Sinkhorn iterations, condition-wise Sinkhorn scales as $O(IBK_g(K_p + K_r))$, while joint Sinkhorn over the expanded

TABLE IV

DIRECT-OUTPUT SWD COMPARISON FOR REFERENCE BASELINES AND MAIN DRIFTING VARIANTS. LOWER IS BETTER. REFERENCE ROWS REPORT MEAN $\pm$ STANDARD DEVIATION FROM THE DIFFUSION/WGAN BENCHMARK OR FROM CHECKPOINT-ONLY DIRECT-SWD EVALUATION. THE DRIFTING-FAMILY ROWS RETAIN DIRECT DRIFTING AND THE TWO SINKHORN TRANSPORT FIELDS USED FOR THE MAIN ABLATION. W-FLOW ROWS REPORT THE AVAILABLE-SEED MEAN $\pm$ STANDARD ERROR. BOLD MARKS THE BEST DRIFTING-FAMILY ROW PER CHANNEL UNDER THE REPORTED MEAN. DASHES MARK UNAVAILABLE MATCHED GENERATOR-LEVEL SWD VALUES.

Method	AWGN	Rayleigh	SSPA	TDL
<i>Diffusion/WGAN reference baselines</i>				
WGAN	0.0198 $\pm$ 0.0050	0.0171 $\pm$ 0.0050	0.0287 $\pm$ 0.0156	0.0289 $\pm$ 0.0063
DDPM	0.0070 $\pm$ 0.0004	0.0079 $\pm$ 0.0008	0.0032 $\pm$ 0.0005	0.0297 $\pm$ 0.0011
DDIM-100	0.0042 $\pm$ 0.0006	0.0044 $\pm$ 0.0007	0.0024 $\pm$ 0.0004	0.0248 $\pm$ 0.0009
<i>Drifting family</i>				
Direct drifting	0.0100 $\pm$ 0.0007	0.0085 $\pm$ 0.0012	0.0060 $\pm$ 0.0008	–
Joint Sinkhorn	0.0148 $\pm$ 0.0002	0.0152 $\pm$ 0.0002	0.0117 $\pm$ 0.0002	0.0966 $\pm$ 0.0005
Condition-wise Sinkhorn	0.0058 $\pm$ 0.0001	0.0073 $\pm$ 0.0001	0.0071 $\pm$ 0.0002	0.0118 $\pm$ 0.0002

batch scales as $O(IB^2 K_g(K_p + K_r))$. At inference, W-Flow, direct drifting, and WGAN retain one generator call per sample, whereas DDPM/DDIM use T denoising-network evaluations.

B. Condition-wise drift-field interpretation

The W-Flow-inspired ablation isolates the drift-field design problem. Joint Sinkhorn drift and condition-wise Sinkhorn drift use the same one-shot generator architecture and differ only in how training particles are moved. Geometrically, joint Sinkhorn builds a transport field from (x, y) -features, while condition-wise Sinkhorn solves separate output-space transport problems at fixed x . This distinction is visible in the mixed metric behavior. Condition-wise Sinkhorn is the strongest W-Flow row under direct SWD on AWGN, Rayleigh, and TDL, while SSPA needs the compact update-budget run because the corrected same-condition field converges earlier than the long diffusion-style preset. Fig. 1 shows the corresponding fixed-condition diagnostic for SSPA. For three transmitted-symbol anchors, the joint Sinkhorn field misses both the conditional mean and the local spread of the analytic output fiber. Condition-wise Sinkhorn keeps the generated cloud close to the analytic same- x cloud. In this channel-simulator setting, the downstream checks below are the more relevant test. Accordingly, global SWD is treated as a comparability diagnostic and supplemented by conditional or downstream metrics for model selection.

C. Conditional metrics and symbolic coding

Table VI reports the corresponding anchor-conditioned diagnostics. The anchor metrics draw repeated outputs at fixed transmitted symbols and therefore probe the conditional law more directly than global SWD. Direct SWD is reported in Table IV, and downstream BER/SER is reported in Table VIII. For AWGN, Rayleigh, SSPA, and TDL, condition-wise Sinkhorn is the strongest evaluated drifting-family surrogate under the downstream BER/SER checks reported here. The full curve comparison below shows that diffusion remains the strongest learned reference on Rayleigh, SSPA, and TDL. On SSPA, the anchor-conditioned metrics also explain

the compact update-budget run. At this budget, the same-condition Sinkhorn field matches fixed-input output clouds well. The full-budget corrected run then degrades the sharp fixed-condition field. The optimizer budget is therefore part of the model selection problem for saturated nonlinear channels.

Table VII makes the SSPA budget selection explicit. The compact condition-wise Sinkhorn result is aggregated over the same 30 seeds used in Tables IV and VIII. The first two rows are single-seed local sanity checks of undertraining. They are not used for the final performance claims. The long full-preset row documents a failure mode of the same corrected field. With fixed drift scale and only $K_g = K_p = K_r = 4$ particles per anchor, the sharp same-condition field becomes a poor long-horizon optimizer unless stopped by conditional diagnostics. After the anchor cloud reaches the analytic floor, continued few-particle updates can over-correct the saturated amplitude law and increase conditional mean and spread errors. The full-preset row improves neither conditional diagnostics nor downstream SER, so the compact row is the reported SSPA operating point.

The downstream symbolic results in Table VIII use the learned channel surrogate for autoencoder training and the analytic channel for final evaluation. The analytic row reports autoencoder training through the analytic channel. Because training and evaluation are finite seeded runs, it can still deviate slightly from an ideal lower-bound interpretation. We treat SER as the primary symbolic metric because the autoencoder message alphabet is finite. BER is retained in the table as an additional bit-level diagnostic. Within the Sinkhorn ablation table, condition-wise Sinkhorn is the more reliable drifting variant and fixes the SSPA failure mode once the update budget is matched to the sharp same-condition field. The per-channel SER curves in Fig. 2 add WGAN and diffusion references to this downstream comparison. At the nominal training point, condition-wise Sinkhorn nearly reaches the analytic AWGN reference, while diffusion is the strongest learned baseline on Rayleigh, SSPA, and TDL. On SSPA, the $M_{\text{msg}} = 64$ update-budget-controlled operating point gives condition-wise Sinkhorn SER 8.13×10^{-5} at $E_b/N_0 = 8$ dB, compared with 7.50×10^{-5} for WGAN, 3.13×10^{-5} for diffusion, and 1.57×10^{-5} for analytic-channel training. The long-block TurboAE check below tests the same AWGN implant outside

TABLE V
PROJECTED TRAINING AND INFERENCE TIMING ON AN NVIDIA RTX 5060 Ti GPU. TRAINING HOURS ARE EXTRAPOLATED FROM A 2% TIMING RUN. INFERENCE IS REPORTED AS TIME PER SAMPLE.

Method	AWGN			Rayleigh			SSPA			TDL		
	Train [h]	Inf.	Total [h]	Train [h]	Inf.	Total [h]	Train [h]	Inf.	Total [h]	Train [h]	Inf.	Total [h]
<i>One-shot generators</i>												
Drifting (dir.)	1.299	0.114 us	1.300	1.299	0.117 us	1.300	5.656	0.118 us	5.656	0.007	0.116 us	0.007
Joint Sinkhorn	0.917	0.116 us	0.918	0.919	0.115 us	0.920	3.956	0.117 us	3.956	0.010	0.116 us	0.010
Condition-wise Sinkhorn	0.048	0.115 us	0.048	0.048	0.115 us	0.049	0.296	0.117 us	0.297	0.009	0.115 us	0.009
WGAN	0.064	0.108 us	0.064	0.085	0.108 us	0.086	0.478	0.107 us	0.478	0.022	0.108 us	0.022
<i>Diffusion samplers</i>												
DDPM	0.020	0.046 ms	0.147	0.020	0.045 ms	0.145	0.124	0.047 ms	0.253	0.005	0.045 ms	0.007
DDIM-100	0.020	0.039 ms	0.128	0.020	0.039 ms	0.128	0.124	0.039 ms	0.231	0.005	0.039 ms	0.006
DDIM-50	0.020	0.019 ms	0.073	0.020	0.019 ms	0.073	0.124	0.019 ms	0.177	0.005	0.019 ms	0.006
DDIM-20	0.020	0.008 ms	0.042	0.020	0.008 ms	0.042	0.124	0.008 ms	0.145	0.005	0.008 ms	0.006
DDIM-10	0.020	0.004 ms	0.031	0.020	0.004 ms	0.031	0.124	0.004 ms	0.135	0.005	0.004 ms	0.005

All rows use the same 2% extrapolation protocol. In the one-shot block, bold total times mark the lowest projected total for each channel. The W-Flow rows use the same one-shot generator architecture as direct drifting. Their training-time differences come only from the drift-field computation.

TABLE VI
ANCHOR-CONDITIONED DIAGNOSTICS FOR SINKHORN DRIFT FIELDS. ANCHOR SWD AND GAUSSIAN WASSERSTEIN-2 (GW2) ARE COMPUTED FROM REPEATED SAMPLES AT FIXED CHANNEL INPUTS. DIRECT SWD AND BER/SER ARE REPORTED SEPARATELY IN TABLES IV AND VIII. VALUES ARE SEED MEANS. LOWER IS BETTER.

Channel	Variant	Anchor SWD	Anchor GW2
AWGN	Joint Sinkhorn	0.1199	0.4443
	Condition-wise Sinkhorn	0.1154	0.4277
Rayleigh	Joint Sinkhorn	0.1723	0.6929
	Condition-wise Sinkhorn	0.1097	0.3820
SSPA	Joint Sinkhorn	0.2174	0.7876
	Condition-wise Sinkhorn	0.0566	0.2376
TDL	Joint Sinkhorn	0.3221	1.1871
	Condition-wise Sinkhorn	0.1054	0.4444

The SSPA condition-wise Sinkhorn row uses the same 30-seed update-budget-controlled run as Table IV.

the short symbolic-codeword setting.

For SSPA, we follow the diffusion-channel setup of [1] more closely and use message alphabet $M_{\text{msg}} = 64$, $n = 8$, rate $6/8$, and training at $E_b/N_0 = 8$ dB. The 30-seed SSPA operating point in Table VIII shows that the corrected condition-wise Sinkhorn implementation is reliable once the W-Flow update budget is matched to the observed convergence of the sharp same-condition field. Fig. 2 reports SER over an E_b/N_0 grid for AWGN, Rayleigh, SSPA, and TDL, which also shows where diffusion retains an advantage over one-shot drifting in downstream coding performance.

D. Equal wall-clock channel-implant training

The previous downstream tables compare final trained channel implants. The equal-wall-clock experiment fixes the symbolic autoencoder training time to 1800 s per implant and records how many optimizer updates fit inside that budget. This directly measures the inner-loop setting that motivates one-shot channel surrogates. DDIM-100 provides a strong learned reference in final-performance comparisons, but its iterative sampler reduces the number of autoencoder updates by

roughly one to two orders of magnitude relative to the one-shot implants. Under this equal wall-clock budget, condition-wise Sinkhorn gives the lowest learned-implant SER on AWGN and SSPA, DDIM-100 is lowest on Rayleigh, and WGAN is lowest on compact TDL. Condition-wise Sinkhorn reaches a competitive downstream operating point while retaining one-shot channel calls, although diffusion or WGAN can remain stronger on individual channels.

E. Long-block TurboAE AWGN check

The symbolic autoencoder results above use short code-words, so we additionally run a longer-block neural-coding check with a CNN TurboAE-style encoder-decoder at block length 64, rate $1/2$, and AWGN training point $E_b/N_0 = 4$ dB. The TurboAE models are trained either through the analytic AWGN channel or through a condition-wise Sinkhorn AWGN surrogate and are then evaluated on the analytic AWGN channel. This experiment complements the short-block multi-channel SER/BER study. It checks whether the learned one-shot channel implant remains usable in a longer differentiable coding loop with a different encoder-decoder architecture.

Fig. 3 shows the resulting BER and BLER curves over 30 seeds. At the 4 dB training point, analytic-channel training gives BER 1.96×10^{-3} with 95% CI 5.21×10^{-4} and BLER 8.18×10^{-2} with 95% CI 2.30×10^{-2} . Training through the condition-wise Sinkhorn surrogate gives BER 3.30×10^{-3} with 95% CI 4.30×10^{-4} and BLER 1.39×10^{-1} with 95% CI 1.91×10^{-2} . The long-block result therefore preserves the waterfall trend and shows successful surrogate-trained TurboAE optimization, with a measurable gap to analytic-channel training.

V. DISCUSSION

A. Accuracy-latency operating point

Condition-wise Sinkhorn drifting targets a low-latency conditional-surrogate operating point. Diffusion remains the strongest learned reference on several hard downstream SER curves and pays for that fidelity through iterative sampling.

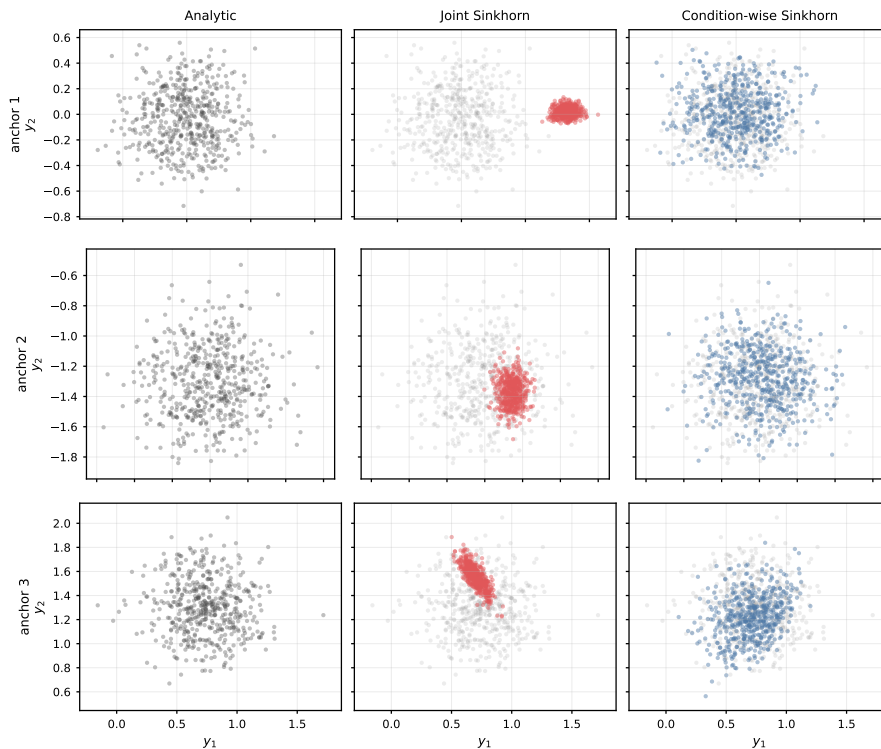


Fig. 1. **Fixed-condition SSPA output fibers.** Each row fixes one transmitted-symbol anchor x_i and plots 512 repeated channel outputs in the first I/Q plane; the three anchors are selected from 256 candidate transmitted symbols. The analytic channel is shown in gray. The learned columns overlay generated samples from joint Sinkhorn and condition-wise Sinkhorn on the same analytic cloud. Joint transport can match aspects of the global output distribution while displacing the fixed- x fiber. Condition-wise Sinkhorn better preserves the conditional output law.

TABLE VII

SSPA CONDITION-WISE SINKHORN BUDGET SENSITIVITY. THE TABLE REPORTS THE CORRECTED CONDITION-WISE SINKHORN FIELD UNDER INCREASING OPTIMIZER BUDGETS. UPDATES ARE APPROXIMATE OPTIMIZER STEPS, COMPUTED FROM DATASET SIZE, BATCH SIZE, AND EPOCHS. RATIOS DIVIDE THE LEARNED-VS-ANALYTIC ANCHOR METRIC BY THE ANALYTIC-VS-ANALYTIC FLOOR. THE FLOOR IS ESTIMATED FROM TWO INDEPENDENT ANALYTIC CHANNEL SAMPLE SETS AT THE SAME ANCHORS. THE FIRST TWO ROWS ARE SINGLE-SEED LOCAL DIAGNOSTICS; THE LAST TWO ROWS ARE AGGREGATED HPC RUNS. LOWER IS BETTER.

Run	Updates	Seeds	Direct SWD	Anchor-SWD ratio	Anchor-GW2 ratio	SER at 8 dB
Short local	0.31k	1	1.07×10^{-1}	5.14	4.10	—
Early compact	0.88k	1	4.09×10^{-2}	4.50	3.88	—
Reported operating point	4.69k	30	$7.07 \times 10^{-3} \pm 1.9 \times 10^{-4}$	1.13 ± 0.005	1.15 ± 0.004	$9.33 \times 10^{-5} \pm 1.4 \times 10^{-5}$
Full preset	390.6k	100	1.53 ± 0.02	35.5 ± 0.4	60.1 ± 0.7	$2.70 \times 10^{-2} \pm 2.2 \times 10^{-3}$

The reported SSPA condition-wise Sinkhorn model is the 4.69k-update operating point selected by anchor-conditioned diagnostics. The full-preset row uses the long diffusion-style SSPA budget with the same corrected field and is included only as a sensitivity diagnostic; it is not used for model selection.

The one-shot drifting path uses a single generator evaluation and preserves the fixed-input structure of the channel law. This makes it relevant when the learned channel is embedded in an inner training loop, where surrogate calls can dominate the computational budget. Diffusion is the safer choice when final standalone fidelity dominates and its sampler cost is acceptable. One-shot Sinkhorn is most attractive when repeated channel calls dominate and the observed SER gap is within tolerance. The main methodological lesson is that channel surrogates should be judged by conditional-law quality and downstream communication metrics in addition to global sample-cloud agreement.

B. Metric-space dependence

The W-Flow run reinforces the metric-dependence of channel-surrogate evaluation. Joint Sinkhorn and condition-wise Sinkhorn can swap order depending on whether the metric is global SWD, anchor-conditioned SWD, or downstream BER/SER. We therefore report direct-output SWD, anchor-conditioned moment metrics, and downstream coding metrics side by side. Hyperparameter selection from a single global SWD score would miss these differences.

C. Conditional Sinkhorn scope

The condition-wise estimator is directly applicable to analytic or simulator channels because repeated outputs can be drawn at the same transmitted symbol. Measured channels

TABLE VIII
DOWNSTREAM SYMBOLIC CODING METRICS FOR SINKHORN CHANNEL SURROGATES. AUTOENCODERS ARE TRAINED THROUGH EACH CHANNEL SURROGATE AND EVALUATED ON THE ANALYTIC CHANNEL. VALUES ARE MEAN $\pm$ STANDARD ERROR OVER SEEDS. LOWER IS BETTER. BOLD MARKS THE BEST LEARNED SURROGATE IN EACH ROW, WHILE THE ANALYTIC CHANNEL IS THE NON-LEARNED REFERENCE TRAINING CONDITION.

Channel	Metric	Analytic	Joint Sinkhorn	Condition-wise Sinkhorn
AWGN	BER	$0.00293 \pm 4.0 \times 10^{-5}$	$0.00388 \pm 5.5 \times 10^{-5}$	$0.00302 \pm 4.2 \times 10^{-5}$
AWGN	SER	$0.00551 \pm 7.3 \times 10^{-5}$	$0.00729 \pm 9.7 \times 10^{-5}$	$0.00568 \pm 7.5 \times 10^{-5}$
Rayleigh	BER	$0.00273 \pm 4.3 \times 10^{-5}$	$0.0177 \pm 3.3 \times 10^{-4}$	$0.00471 \pm 8.2 \times 10^{-5}$
Rayleigh	SER	$0.00514 \pm 8.0 \times 10^{-5}$	$0.0331 \pm 5.6 \times 10^{-4}$	$0.00881 \pm 1.4 \times 10^{-4}$
SSPA	BER	$6.72 \times 10^{-6} \pm 1.9 \times 10^{-6}$	–	$4.90 \times 10^{-5} \pm 7.5 \times 10^{-6}$
SSPA	SER	$1.30 \times 10^{-5} \pm 3.5 \times 10^{-6}$	–	$9.33 \times 10^{-5} \pm 1.4 \times 10^{-5}$
TDL	BER	$0.00158 \pm 2.1 \times 10^{-5}$	$0.0107 \pm 3.7 \times 10^{-4}$	$0.00293 \pm 4.4 \times 10^{-5}$
TDL	SER	$0.00297 \pm 3.6 \times 10^{-5}$	$0.02 \pm 6.3 \times 10^{-4}$	$0.00546 \pm 7.6 \times 10^{-5}$

SSPA uses the 30-seed $M_{\text{msg}} = 64$ update-budget-controlled operating point. Dashes mark variants not rerun under that SSPA coding setup.

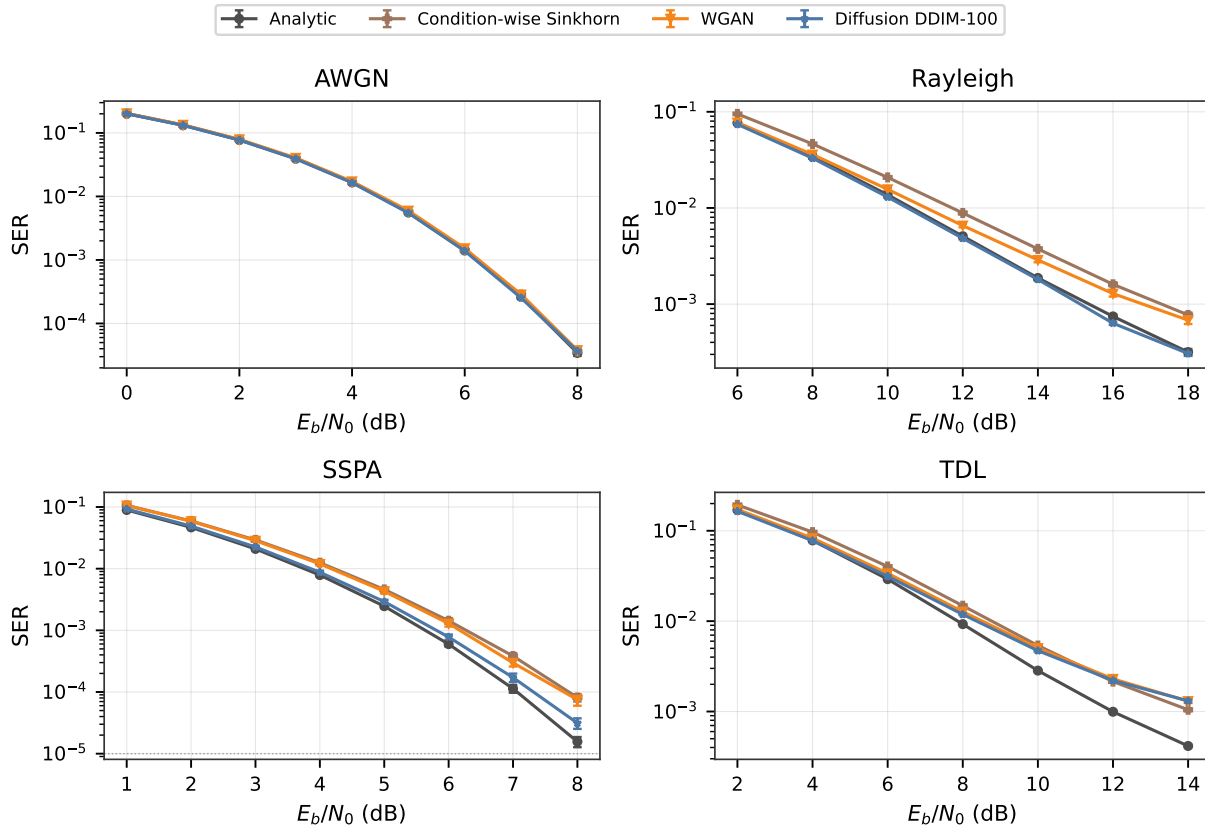


Fig. 2. **SER curves for learned channel surrogates.** Symbolic autoencoders are trained through each surrogate at the nominal training point and evaluated on the analytic channel over an E_b/N_0 grid for AWGN, Rayleigh, SSPA, and TDL. Curves report seed means with standard-error bars; points below the single-run Monte Carlo resolution are clipped to the dotted reporting floor and marked as upper bounds.

are harder. With only one observation per x , the exact conditional law $p(\cdot | x)$ cannot be observed at a point. For that regime, the local conditional Sinkhorn approximation in Sec. II estimates the target fixed-input output law from kernel-weighted neighborhoods in condition space and still solves the transport problem only in the output coordinate. This keeps the fixed-input geometry intact, but introduces bandwidth selection and neighborhood bias. The present experiments do not evaluate this measured-data approximation. Validating it for measured-channel data is left for future work. Practical measured-channel simulators may also need transport features

that encode phase, amplitude, input-output geometry, channel-state information, or decoder-relevant projections.

VI. CONCLUSION

This paper makes one-shot drifting applicable to learned channel simulation by replacing unconditional transport with condition-wise Sinkhorn transport over fixed-input conditional output laws. The resulting generator preserves the transmitted symbol x , transports only the conditional output law, and keeps the one-shot inference path that motivates drifting for inner-loop channel use. The population formulation identifies

TABLE IX

EQUAL WALL-CLOCK SYMBOLIC AUTOENCODER TRAINING. EACH ROW TRAINS THE SAME SYMBOLIC AUTOENCODER FOR 1800 s THROUGH THE LISTED CHANNEL IMPLANT AND EVALUATES THE RESULTING ENCODER/DECODER ON THE ANALYTIC CHANNEL. RESULTS ARE MEAN $\pm$ STANDARD ERROR OVER 30 SEEDS.

Channel	Training implant	Updates [k]	SER	BER
AWGN	Analytic	705.7	$3.54 \times 10^{-3} \pm 3.94 \times 10^{-5}$	$1.89 \times 10^{-3} \pm 2.31 \times 10^{-5}$
	Condition-wise Sinkhorn	626.8	$3.62 \times 10^{-3} \pm 3.65 \times 10^{-5}$	$1.93 \times 10^{-3} \pm 2.31 \times 10^{-5}$
	WGAN	641.8	$3.70 \times 10^{-3} \pm 4.06 \times 10^{-5}$	$1.98 \times 10^{-3} \pm 2.87 \times 10^{-5}$
	DDIM-100	21.5	$5.32 \times 10^{-3} \pm 1.36 \times 10^{-4}$	$2.83 \times 10^{-3} \pm 6.82 \times 10^{-5}$
Rayleigh	Analytic	656.3	$2.25 \times 10^{-3} \pm 4.57 \times 10^{-5}$	$1.22 \times 10^{-3} \pm 2.94 \times 10^{-5}$
	Condition-wise Sinkhorn	602.1	$5.40 \times 10^{-3} \pm 1.11 \times 10^{-4}$	$2.92 \times 10^{-3} \pm 7.00 \times 10^{-5}$
	WGAN	614.0	$4.83 \times 10^{-3} \pm 2.87 \times 10^{-4}$	$2.55 \times 10^{-3} \pm 1.52 \times 10^{-4}$
	DDIM-100	21.3	$4.64 \times 10^{-3} \pm 1.08 \times 10^{-4}$	$2.47 \times 10^{-3} \pm 6.10 \times 10^{-5}$
SSPA	Analytic	635.6	$1.33 \times 10^{-6} \pm 6.31 \times 10^{-7}$	$7.78 \times 10^{-7} \pm 3.81 \times 10^{-7}$
	Condition-wise Sinkhorn	617.1	$8.80 \times 10^{-6} \pm 1.87 \times 10^{-5}$	$4.36 \times 10^{-5} \pm 8.30 \times 10^{-6}$
	WGAN	627.3	$9.83 \times 10^{-5} \pm 3.15 \times 10^{-5}$	$4.50 \times 10^{-5} \pm 1.27 \times 10^{-5}$
	DDIM-100	21.4	$9.00 \times 10^{-5} \pm 4.64 \times 10^{-5}$	$4.03 \times 10^{-5} \pm 1.72 \times 10^{-5}$
TDL	Analytic	449.0	$1.30 \times 10^{-3} \pm 1.98 \times 10^{-5}$	$6.98 \times 10^{-4} \pm 1.19 \times 10^{-5}$
	Condition-wise Sinkhorn	576.8	$3.44 \times 10^{-3} \pm 6.05 \times 10^{-5}$	$1.82 \times 10^{-3} \pm 3.43 \times 10^{-5}$
	WGAN	587.6	$3.21 \times 10^{-3} \pm 4.36 \times 10^{-4}$	$1.70 \times 10^{-3} \pm 2.19 \times 10^{-4}$
	DDIM-100	21.3	$4.64 \times 10^{-3} \pm 1.23 \times 10^{-4}$	$2.47 \times 10^{-3} \pm 6.36 \times 10^{-5}$

Bold marks the best learned implant per channel; the analytic row is the non-learned reference. The DDIM row uses 100 denoising steps per channel call, which yields many fewer optimizer updates within the same autoencoder-training time budget.

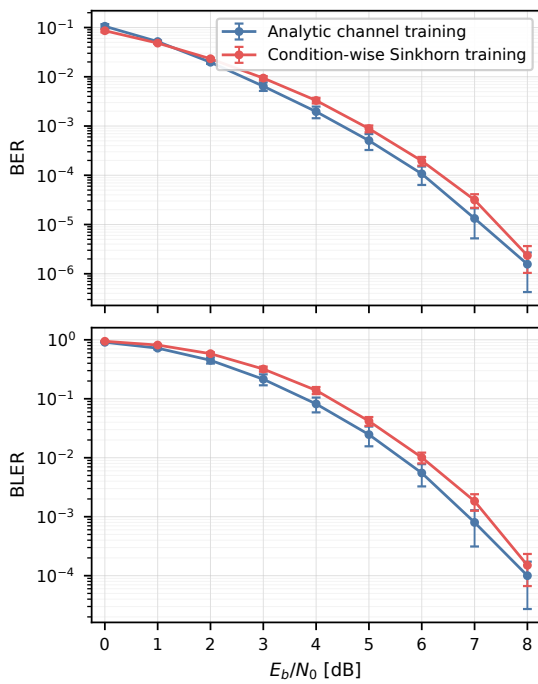


Fig. 3. **Long-block TurboAE AWGN check.** CNN TurboAE models at block length 64, rate 1/2, are trained through either analytic AWGN or the condition-wise Sinkhorn AWGN surrogate and evaluated on analytic AWGN over 30 seeds. The surrogate preserves the waterfall trend with a measurable gap to analytic-channel training.

equality of $p(\cdot | x)$ and $q_\theta(\cdot | x)$ for μ -almost every x as the correct equilibrium geometry. The implemented method is a finite-sample barycentric approximation followed by projected detached-target neural training.

Experiments on AWGN, Rayleigh, SSPA, and compact TDL channels show a clear accuracy-latency tradeoff. Within the matched Sinkhorn drift-field ablation, condition-wise

Sinkhorn is the strongest evaluated one-shot surrogate under the downstream symbolic-coding checks. Diffusion remains the strongest learned reference on the hardest SER curves. This positions the contribution as a low-latency, condition-preserving surrogate for settings where repeated channel calls are expensive. The SSPA study also shows that optimizer budget is part of the drift-field design. The corrected same-condition field works well after convergence, and the full diffusion-style preset can degrade this sharper one-step field. Across the full set of results, global SWD, anchor-conditioned metrics, and downstream coding metrics can disagree. This makes metric choice a central part of learned channel simulation and motivates future work on channel-specific transport features and local conditional Sinkhorn approximations for measured channels.

REFERENCES

- [1] M. Kim, R. Fritschek, and R. F. Schaefer, “Diffusion models for accurate channel distribution generation,” *arXiv preprint arXiv:2309.10505*, 2023.
- [2] —, “Robust generation of channel distributions with diffusion models,” in *ICC 2024 – IEEE International Conference on Communications*, 2024, pp. 330–335.
- [3] T. Lee, J. Park, H. Kim, and J. G. Andrews, “Generating high dimensional user-specific wireless channels using diffusion models,” *IEEE Transactions on Wireless Communications*, vol. 25, pp. 2907–2921, 2026.
- [4] X. Gong, X. Liu, A. A. Lu, X. Gao, X. G. Xia, C.-X. Wang, and X. You, “Digital twin of channel: Diffusion model for sensing-assisted statistical channel state information generation,” *IEEE Transactions on Wireless Communications*, vol. 24, no. 5, pp. 3805–3821, 2025.
- [5] X. Zhou, L. Liang, J. Zhang, P. Jiang, Y. Li, and S. Jin, “Generative diffusion models for high dimensional channel estimation,” *IEEE Transactions on Wireless Communications*, vol. 24, no. 7, pp. 5840–5854, 2025.
- [6] B. Fesl, M. Baur, F. Strasser, M. Joham, and W. Utschick, “Diffusion-based generative prior for low-complexity MIMO channel estimation,” *IEEE Wireless Communications Letters*, vol. 13, no. 12, pp. 3493–3497, 2024.
- [7] Z. Chen, H. Shin, and A. Nallanathan, “Generative diffusion model-based variational inference for MIMO channel estimation,” *IEEE Transactions on Communications*, vol. 73, no. 10, pp. 9254–9269, 2025.

- [8] N. Zilberstein, A. Swami, and S. Segarra, "Joint channel estimation and data detection in massive MIMO systems based on diffusion models," in *ICASSP 2024 – IEEE International Conference on Acoustics, Speech and Signal Processing*, 2024, pp. 13 291–13 295.
- [9] H. Fu, W. Si, and R. Liu, "Conditional denoising diffusion-based channel estimation for fast time-varying MIMO-OFDM systems," *Digital Signal Processing*, vol. 164, p. 105283, 2025.
- [10] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 6840–6851.
- [11] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," in *International Conference on Learning Representations*, 2021.
- [12] Y. Song, P. Dhariwal, M. Chen, and I. Sutskever, "Consistency models," in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202. PMLR, 2023, pp. 32 211–32 252. [Online]. Available: <https://proceedings.mlr.press/v202/song23a.html>
- [13] T. Yin, M. Gharbi, R. Zhang, E. Shechtman, F. Durand, W. T. Freeman, and T. Park, "One-step diffusion with distribution matching distillation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2024, pp. 6613–6623.
- [14] M. Deng, H. Li, T. Li, Y. Du, and K. He, "Generative modeling via drifting," *arXiv preprint arXiv:2602.04770*, 2026.
- [15] M. Sahraee-Ardakan, M. Delbracio, and P. Milanfar, "The geometry of noise: Why diffusion models don't need noise conditioning," *arXiv preprint arXiv:2602.18428*, 2026.
- [16] J. Han, P. Li, Q. Guo, R. Xu, S. Ermon, and E. J. Candès, "One-step generative modeling via Wasserstein gradient flows," *arXiv preprint arXiv:2605.11755*, 2026.
- [17] M. Cuturi, "Sinkhorn distances: Lightspeed computation of optimal transport," in *Advances in Neural Information Processing Systems*, vol. 26, 2013, pp. 2292–2300.
- [18] J. Feydy, T. Séjourné, F.-X. Vialard, S.-i. Amari, A. Trounev, and G. Peyré, "Interpolating between optimal transport and MMD using Sinkhorn divergences," in *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 89. PMLR, 2019, pp. 2681–2690. [Online]. Available: <https://proceedings.mlr.press/v89/feydy19a.html>
- [19] 3GPP, "Study on Channel Model for Frequencies from 0.5 to 100 GHz," 3GPP, Technical Report TR 38.901, 2022, version 17.1.0.